Lesson notes · for you

Pack the truck

Optimization · Level 1 · Lesson 1

Time	Materials
20 min	Paper and a pen; Six small things from your bag, or six paper cards

Learning goals

- Say the rule and the goal of a packing puzzle.
- Check whether a load keeps the rule.
- Find the best load, and think about what stays behind.

Activity: Pack the bag

1. Cards (5 min): Write six things you might carry, such as a book, a water bottle, lunch, a jumper, a ball and a pencil case. Give each a weight from 1 to 5 and points from 1 to 10.
2. Pack (8 min): Put cards together and add up their weights. Any load that adds up to more than 10 doesn't count. Write down three loads that fit.
3. Best (5 min): Add up the points of each load that fits. Circle the one with the most points.
4. Share (2 min): Show your partner your best load. Can they find one with more points that still fits?

No cards? Tear a page into six pieces. No device needed.

Discussion

- Why didn't the quick way give the best load?
- Why doesn't a load that is too heavy count?
- Who should decide what stays behind?

Check yourself

- The best truck load is ___ and ___, with ___ points.

Try it this week

Pack your school bag like a truck. Give each thing a weight (1 to 5) and points (how much you need it), and choose a limit, like 10. Find the load with the most points that stays under the limit, then ask a friend or someone at home to beat it.

Worksheet

Pack the truck

Optimization · Level 1 · Lesson 1

Name _____ Class _____ Date _____

1 **You load the desks (500 kg, 33 points) and the water filter (300 kg, 15 points): 800 kg and 48 points. Is it the best load?**

◯ Yes: it keeps the rule ◯ No: desks and books give more points ◯ No: it's too heavy

2 **A load that is too heavy can still be the best load if it has the most points.**

◯ True ◯ False

3 **Match each word to what it means in the truck puzzle.**

The rule	• • • •	The six loads
The goal	• • • •	No more than 1,000 kg
The choices	• • • •	The most points

4 **Which loads fit on the truck together (1,000 kg or less)? Choose all that fit.**

☐ Solar panels, books and chalk (600 + 500 + 100) ☐ Solar panels and desks (600 + 500)

☐ Desks and books (500 + 500) ☐ Books, water filter and footballs (500 + 300 + 200)

☐ Solar panels and the water filter (600 + 300)

5 **Put the loads in order, from the most points to the fewest.**

☐ Water filter (15)

☐ Books (32)

☐ Solar panels (42)

☐ Desks (33)

☐ Chalk and paper (3)

☐ Footballs (8)

✎ **Think of a time you couldn't take everything: a bag, a trip, pocket money. What was your rule, what was your goal, and what stayed behind?**

Check your answers

Pack the truck

Optimization · Level 1 · Lesson 1

1 **No: desks and books give more points**

Your load keeps the rule (800 kg is under 1,000 kg), so it is allowed.

2 **False**

The rule comes first. A load over 1,000 kg can't go on the truck at all, so it can't be the best, however many points it has.

3 **The rule — No more than 1,000 kg**
The goal — The most points
The choices — The six loads

You choose from the loads, you must keep the weight rule, and you want the most points.

4 **Desks and books (500 + 500); Solar panels and the water filter (600 + 300); Books, water filter and footballs (500 + 300 + 200)**

Add the weights: desks and books make 1,000 kg, solar and filter 900 kg, and books, filter and footballs 1,000 kg: all fit.

5 **Top to bottom: 4, 3, 1, 2, 6, 5**

The solar panels have the most points, but the most points alone isn't the whole story: they are also the heaviest, and taking them leaves room for less.